Severity Based Fire Detection using Computer Vision

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***Abstract*—Modern computer vision systems for fire detection are frequently compromised by the "context problem," where benign ignition sources like candles trigger false emergency alerts. This paper presents a conceptual optimization strategy to resolve this operational barrier. Moving beyond traditional "presence-based" detection, we implemented a Divergent Bimodal Approach using a YOLOv8s architecture. Initially, a granular 1–5 severity scale was tested but found to be impractical due to feature convergence and "mid-range blur." By pivoting to two distal anchors: $fire$ (contained) and $HZRD$ (hazardous) and integrating structural cues and smoke patterns, we maximized the latent distance between classes. Our results on a 1,514-image dataset show that anchoring detection at these extremes provides the situational awareness necessary for reliable real-world deployment. This methodology offers a computationally efficient blueprint for integrating smart fire safety into existing CCTV infrastructure.**

***Index Terms*—Computer vision, fire detection, YOLOv8, bimodal classification, structural cues.**

I. Introduction

Conventionally, fire detection infrastructure has been centered around physical sensors (heat and smoke). However, these require comprehensive, dedicated systems that are often costly and rely heavily on physical constraints, such as enclosed environments, to "trap" particulates. Consequently, they often trigger only once a hazard has reached a "point of no return".

Computer Vision (CV) offers a proactive alternative. It provides versatility by monitoring open-air environments like forests where traditional sensors are ineffective. Modern architectures like YOLOv8s are highly efficient and, while capable of running on "edge" devices, are most reliably integrated on the server-side of existing CCTV networks. Furthermore, CV reduces latency by identifying combustion at the "visual ignition" stage, often minutes before smoke reaches a physical sensor.

A. Core Issue with CV

Despite these advantages, CV fire detection has struggled with Operational Feasibility. The primary barrier is not the detection of "orange pixels," but their semantic interpretation. To a standard AI model, a candle flame and a car fire are nearly identical clusters of high-intensity color. This lack of context leads to two critical failures:

- *False Positive Fatigue:* Systems trigger emergency alerts for harmless, controlled sources such as lighters or candles.
- *Environmental Mimicry:* Visual "noise," such as sunsets or red vehicle reflections, is often misidentified as a hazard.

B. Approaches in Related Studies

Recent studies have attempted to bridge this gap through several Resource-Intensive Optimizations. One prominent approach is double-layer verification, utilizing post-detection checks or multi-sensor Bayesian decision fusion to refine model confidence [2], [5]. Other researchers have focused on spatial and temporal video analysis, investigating flickering patterns and motion features to verify fire regions [1], [4]. Furthermore, the incorporation of structural cues, such as tree patterns and smoke plumes, has been explored to assist models in early detection [3], [6]. Finally, many studies rely on massive data collection and architectural modifications to the YOLO backbone to force higher accuracy [7]. While these methods improve raw detection quality, they rely on more data or processing layers. This paper proposes an optimization at the conceptual level.

II. Methodology

A. Model Selection and Architectural Justification

The YOLOv8s (You Only Look Once, version 8 small) architecture was selected as the primary backbone for this research. This choice was driven by the necessity for computational efficiency and generalization capability. Unlike larger architectures such as YOLOv11l, which pose a high risk of memorizing specific pixel noise or backgrounds on a relatively small dataset (1,514 images), the YOLOv8s variant provides sufficient depth to learn underlying semantic patterns without overfitting. The comparison of model complexity is detailed in Table I, illustrating why the "small" variant represents an optimal balance for server-side integration on standard hardware.

B. Preliminary Study: Scale Based Grading

Before finalizing the architecture, a preliminary feasibility study utilizing a Linear Grading Scale (labeled $fire\_1$ through $fire\_5$) was conducted. The objective was to determine if a

TABLE I
COMPARISON OF MODEL COMPLEXITY AND PARAMETERS

Model Variant	Parameters (M)	GFLOPs
YOLOv8n (Nano)	3.2	8.7
YOLOv8s (Small)	**11.2**	**28.6**
YOLOv8m (Medium)	25.9	78.9
YOLOv11l (Large)	54.1	150+

lightweight model could provide high-resolution severity data by categorizing flames into five distinct intensity levels. While conceptually attractive, qualitative analysis of the training cycles revealed a critical bottleneck: the visual signatures of mid-range severity levels were found to be too similar for the feature extraction layers to distinguish consistently. This created a "mid-range blur" in the latent space, where the model struggled to differentiate between levels such as $fire\_2$ and $fire\_3$, maintaining the high false-positive rates the system aimed to eliminate.

C. Dual-Anchor Severity Mapping

Based on the results of the preliminary study, the methodology was pivoted toward Dual-Anchor Severity Mapping. Instead of a continuous spectrum, the logic was re-engineered to focus on two distal anchors: $fire$ (Contained), representing localized, safe ignition sources like matchsticks or lighters, and $HZRD$ (Hazardous), representing high-energy, uncontrolled combustion like violent vehicle or forest fires. By training the model on these extremes, we maximize the latent distance between classes and force a categorical decision that aligns with human emergency judgment. This polarization allows the model to utilize structural cues—recognizing that threat levels are defined by the context of ignition (e.g., a forest floor vs. a kitchen stove) rather than simple pixel intensity.

III. FINDINGS

The transition from scale-based grading to a binary class model resulted in a significant stabilization of decision boundaries. As shown in Table II, the bimodal anchoring approach achieved an mAP@50 score of 64%, a substantial improvement over the 53% achieved by the granular grading scale. This quantitative jump suggests that the historical failure of fire safety CV is often a lack of semantic intent rather than image resolution.

TABLE II
COMPARATIVE PERFORMANCE OF DETECTION METHODOLOGIES

Method	mAP@50 Score (%)
Scale Based Grading	53
Binary Class Model (Proposed)	**64**

Qualitative results in Fig. 1 demonstrate the efficacy of this method. Testing showed that the model successfully utilized environmental context for classification. For instance, a small flame detected within the region of a kitchen stove was correctly ignored as $fire$, whereas a flame of the same pixel-scale detected within a vehicle chassis or accompanied by black smoke was flagged as $HZRD$. While large bonfires remain a primary source of classification errors, this "over-detection" is a safer failure mode for public safety than previous ambiguity.

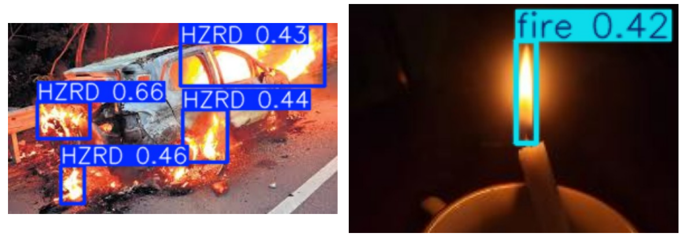

Fig. 1. Model inference: Hazardous vehicle fire (left) vs. contained candle fire (right).

IV. CONCLUSION

This research provides a practical blueprint for making fire detection a feasible utility in existing infrastructure. By moving away from "presence-only" detection and adopting a severity-based bimodal approach, we have addressed the context problem that leads to high false-alarm rates. This logic remains computationally efficient through the YOLOv8s backbone, offering a path for server-side integration into CCTV networks without massive hardware investment. Future improvements will focus on integrating smoke as a separate class to further harden the system against environmental mimicry. Ultimately, a conceptually optimized model can achieve the level of situational awareness required for trusted, autonomous emergency response.

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